The Distributive Property

Lesson 6-5

Name: ____________ Date: ____________ Class: ____________

Key Vocabulary

Level 1 support

Picture first, then the word, then a plain-language meaning. Say each word out loud.

$3(4 + 5) = 3×4 + 3×5 = 12 + 15 = 27$ — the 3 gets 'distributed' to both the 4 and the 5

Distributive Property

Multiplying a number by everything inside the parentheses: $a(b + c) = ab + ac$.

In $3(x + 2)$, the 3 is the factor outside the parentheses that multiplies each term inside

Factor

A number that gets multiplied by another number.

Expand $2(n + 6)$: multiply $2 × n = 2n$ and $2 × 6 = 12$, so $2(n + 6) = 2n + 12$

Expand

To multiply out the parentheses in an expression.

$3(x + 4)$ and $3x + 12$ always give the same answer: when $x = 2$, both $= 18$; when $x = 10$, both $= 42$

Equivalent

Expressions that always have the same value.

In $6x + 15$, the coefficient of x is 6 — it came from distributing in $3(2x + 5)$

Coefficient

The number in front of a letter, like the 3 in $3x$.

$4x + 2x = 6x$ (combine coefficients); $4x + 2y$ cannot be combined (different variables)

Like terms

Terms with the same letter, like $2x$ and $5x$.

Key Ideas & Notes

- You're selling concert bundles at the music studio.
- Each bundle includes a ticket ($15) and a snack ($5).
- If 3 people each buy a bundle, you can calculate the total as $3(15 + 5)$ OR as $3 \times 15 + 3 \times 5$.
- Either way, the total is the same — $60!
- Expand each expression using the distributive property. Then simplify to find the value.

Think About It

- Why do both methods give the same total?
- What does the 3 represent in $3(15 + 5)$?
- How is $3(15 + 5)$ related to $3 \times 15 + 3 \times 5$?

My Notes

Turn & Talk — Discussion Points

Level 1 support | Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

LAUNCH

Each bundle is a ticket ($15) and a snack ($5). For 3 people, why does $3(15 + 5)$ give the same $60 as $3 \times 15 + 3 \times 5$?

Level 1 support

Start here: The distributive property multiplies the 3 by each item, giving the same total.

Both give $60 because I distribute the 3 to ____ and ____.

Las dos dan $60 porque distribuyo el 3 a ____ y ____.

So $3(15 + 5)$ equals ____ because ____.

Entonces $3(15 + 5)$ es igual a ____ porque ____.

Word bank: Distributive Property | Factor | Expand | Equivalent

Listen for: *Students explain the 3 distributes to both the ticket and the snack ($3 \times 15 + 3 \times 5 = 45 + 15 = 60$), matching 3×20, and use the word equivalent.*

Level 2

If 10 people bought bundles, which form is faster to compute: $10(15 + 5)$ or $10 \times 15 + 10 \times 5$? Justify your choice.

- The ____ form is faster because ____.
- Either way the total is ____, since both forms are equivalent because ____.

EXPLORE

When you expand 5(2x + 4), why must the 5 multiply BOTH terms inside the parentheses?

Level 1 support

Start here: The factor outside multiplies every term inside the parentheses.

I distribute the 5 to ___ and ___, so 5(2x + 4) = ___.

Distribuyo el 5 a ___ y ___, así 5(2x + 4) = ___.

Both terms get multiplied because ___.

Los dos términos se multiplican porque ___.

Word bank: **Distributive Property** **Expand** **Factor** **Equivalent**

Listen for: *A strong answer expands 5(2x + 4) = 10x + 20 and explains the parentheses mean 5 copies of the whole sum, so both 2x and 4 are multiplied.*

Level 2

A classmate writes 5(2x + 4) = 10x + 4. Critique the error and prove the correct form is equivalent by testing x = 3.

- The error is ___ because they only multiplied ___.
- Testing x = 3: 5(2·3 + 4) = ___ but 10·3 + 4 = ___, so they are not equivalent.

CONNECT

For 8 practice rooms each needing a music stand ($12) and a stool ($9), the cost is 8(12 + 9). How do you use the distributive property to find the total, and which form is easier for you?

Level 1 support

Start here: Distribute the 8 to both items, then add.

$8(12 + 9) = 8 \times$ ____ $+ 8 \times$ ____ $=$ ____.

$8(12 + 9) = 8 \times$ ____ $+ 8 \times$ ____ $=$ ____.

I think the ____ form is easier because ____.

Creo que la forma ____ es más fácil porque ____.

Word bank: **Distributive Property** **Factor** **Expand** **Equivalent**

Listen for: *Students compute $8 \times 12 + 8 \times 9 = 96 + 72 = 168$ (or $8 \times 21 = 168$) and justify which form they find easier, recognizing both are equivalent.*

Level 2

Working backwards, how would you factor 8x + 12 into a product? Explain how factoring undoes the distributive property.

- I factor out the ____, giving ____.
- Factoring undoes expanding because ____.

Guided Examples

Example 1

Expand $6(n + 3)$ using the distributive property.

Solution: $6(n + 3) = 6 \times n + 6 \times 3 = 6n + 18$.

Answer: A. $6n + 18$

Example 2

Expand $4(5 - 2)$ using the distributive property.

Solution: $4(5 - 2) = 4 \times 5 - 4 \times 2 = 20 - 8 = 12$.

Answer: A. 12

Example 3

Expand $3(x + 4)$ using the distributive property.

Solution: $3(x + 4) = 3 \times x + 3 \times 4 = 3x + 12$.

Answer: A. $3x + 12$

Write About the Math

The Writing Revolution

I can explain my steps using the words distributive property, factor, expand, and equivalent.

1. Kernel Sentence

subject + verb

Model: Distributive Property is multiplying a number by everything inside the parentheses: $a(b + c) = ab + ac$.

Propiedad distributiva es multiplicar un número por todo lo que está dentro del paréntesis: $a(b + c) = ab + ac$.

Write a kernel sentence about distributive property. Use a subject and a verb.

Escribe una oración base sobre propiedad distributiva. Usa un sujeto y un verbo.

2. Sentence Expansion

because · but · so

Kernel: Distributive Property matters in math

Propiedad distributiva importa en matemáticas

Expand the kernel three ways. Add a reason, a contrast, and a result.

because *porque*	**Distributive Property matters in math because ____.** *Propiedad distributiva importa en matemáticas porque ____.*
but *pero*	**Distributive Property matters in math, but ____.** *Propiedad distributiva importa en matemáticas, pero ____.*
so *entonces*	**Distributive Property matters in math, so ____.** *Propiedad distributiva importa en matemáticas, entonces ____.*

3. Sentence Types

4 ways to write a math idea

Statement
Afirmación

Tell one true fact about distributive property.
Di un hecho verdadero sobre distributive property.

Distributive property ____.

Question
Pregunta

Ask a question about distributive property.
Haz una pregunta sobre distributive property.

How does ____ ?
¿Cómo ____ ?

Exclamation
Exclamación

Show excitement about distributive property.
Muestra entusiasmo sobre distributive property.

Wow, ____ !
¡Guau, ____ !

Command
Mandato

Tell a partner what to do with distributive property.
Dile a un compañero qué hacer con distributive property.

First, ____ .
Primero, ____ .

4. Explain Your Reasoning

use a sentence starter

Both give $60 because I distribute the 3 to ____ and ____.
Las dos dan $60 porque distribuyo el 3 a ____ y ____.

So 3(15 + 5) equals ____ because ____.
Entonces 3(15 + 5) es igual a ____ porque ____.

I distribute the 5 to ____ and ____, so 5(2x + 4) = ____.
Distribuyo el 5 a ____ y ____, así 5(2x + 4) = ____.

Try It

Solve on your own. Check the answer key when you are done.

1. Expand 3(x + 4) using the distributive property.

A. 3x + 12

B. 3x + 4

C. x + 12

D. 7x

Show your work:

2. When using the distributive property with a(b + c), you must multiply 'a' by:

A. Both b and c

B. Only b

C. Only c

D. b + c first, then multiply

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

Use the distributive property to explain why 6 × 98 can be calculated as 6 × 100 − 6 × 2 = 588. Then create your own example of using the distributive property to make multiplication easier.

Sentence starter: 6 × 98 = 6(100 − 2) = 6 × ____ − 6 × ____ = ____ − ____ = ____. My example: ____ × ____ = ____(____ ± ____) = ____.

Show your work:

Reflect — Exit Ticket

Which expression is equivalent to 7(x + 3)?

A. 7x + 21

B. 7x + 3

C. x + 21

D. 7x + 10

Your answer:

Answer Key & Teacher Guide

1. **Try It 1:** A. 3x + 12 *— 3(x + 4) = 3 × x + 3 × 4 = 3x + 12.*
2. **Try It 2:** A. Both b and c *— The distributive property says a(b + c) = ab + ac. You multiply the outside factor by EACH term inside the parentheses.*
3. **Exit Ticket:** A. 7x + 21 *— 7(x + 3) = 7 × x + 7 × 3 = 7x + 21.*

Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Distributive Property is multiplying a number by everything inside the parentheses: a(b + c) = ab + ac.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).